

# MENTORA ACADEMY

EXCELLENCE IN EDUCATION

## Edexcel International A Level Chemistry (WCH14)

### UNIT 4 — MASTER TOPICAL PRACTICE PACK: 12B

#### LATTICE ENERGY, BORN-HABER CYCLES & SOLUTION ENERGETICS

|                            |                                                              |
|----------------------------|--------------------------------------------------------------|
| <b>CANDIDATE NAME:</b>     | Usman                                                        |
| <b>SUBJECT &amp; CODE:</b> | Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)       |
| <b>TOPIC AREA:</b>         | Topic 12: ENTROPY AND ENERGETICS                             |
| <b>SUB-TOPIC:</b>          | 12B: Lattice Energy, Born-Haber Cycles & Solution Energetics |
| <b>TOTAL QUESTIONS:</b>    | 50 Authentic Past Paper Questions (Tier 1 & Tier 2)          |
| <b>TOTAL MARKS:</b>        | 233 Marks                                                    |
| <b>EXAMINER FAQs:</b>      | 10 Empirical Examiner Traps & Model Answers Included         |

#### HOW TO USE THIS A\* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A\* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

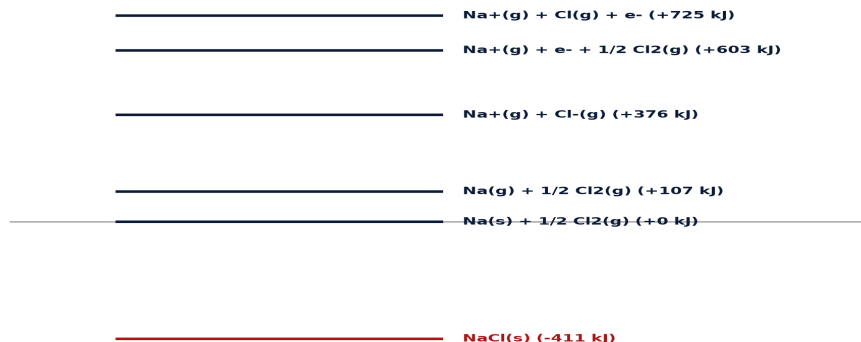
## SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25

## Q1. 1. Lattice Energy Definition

[3 Marks]  
WCH14/01/Jan23/Q7

Lattice energy is a measure of the strength of ionic bonding in a crystal lattice.

## Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Define standard lattice energy ( $\Delta_{\text{LE}} \text{H}^\circ$ ). [2]

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(b) Explain why lattice energy values are always negative. [1]

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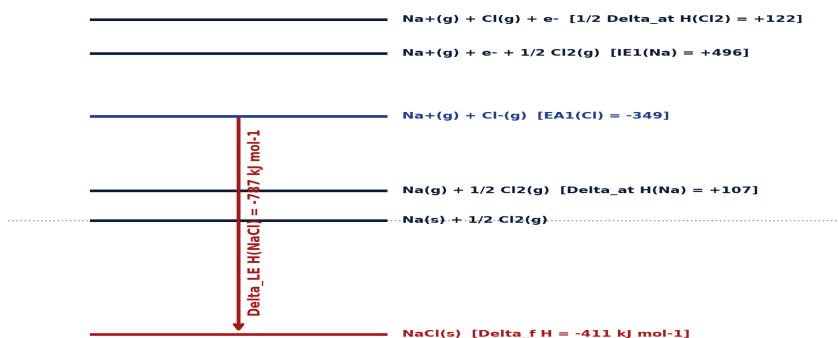
## Q2. 2. Born-Haber Cycle Calculation for NaCl

[5 Marks]  
WCH14/01/Oct22/Q8

Consider the following standard enthalpy changes ( $\text{kJ mol}^{-1}$ ):

$\Delta_{\text{f}} \text{H}(\text{NaCl}) = -411$ ,  $\Delta_{\text{at}} \text{H}(\text{Na}) = +107$ ,  $\text{IE1}(\text{Na}) = +496$ ,  $\frac{1}{2} \Delta_{\text{at}} \text{H}(\text{Cl}_2) = +122$ ,  $\text{EA1}(\text{Cl}) = -349$ .

## Born-Haber Cycle for Sodium Chloride, NaCl(s)



(a) Construct the Born-Haber cycle for NaCl(s) and calculate the lattice energy  $\Delta_{\text{LE}} \text{H}(\text{NaCl})$ . [4]

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(b) State how lattice energy relates to the melting point of NaCl. [1]

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### Q3. 3. First Electron Affinity Definition & Sign

[3 Marks]  
WCH14/01/Jun22/Q9

The first electron affinity of chlorine is  $EA_1 = -349 \text{ kJ mol}^{-1}$ .

(a) Define first electron affinity. [2]

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(b) Explain why the first electron affinity of chlorine is exothermic. [1]

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### Q4. 4. Second Electron Affinity Endothermic Nature

[3 Marks]  
WCH14/01/Jan22/Q10

The second electron affinity of oxygen is  $EA_2(O) = +798 \text{ kJ mol}^{-1}$ .

(a) Write an equation representing the second electron affinity of oxygen. [1]

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(b) Explain why the second electron affinity is endothermic while the first is exothermic. [2]

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### Q5. 5. Lattice Energy Comparison: NaCl vs MgCl<sub>2</sub>

[4 Marks]  
WCH14/01/Oct21/Q8

$\Delta_{LE} H(\text{NaCl}) = -787 \text{ kJ mol}^{-1}$ , whereas  $\Delta_{LE} H(\text{MgCl}_2) = -2525 \text{ kJ mol}^{-1}$ .

#### Born-Haber Cycle for Sodium Chloride (NaCl)

|       |                                                                                          |
|-------|------------------------------------------------------------------------------------------|
| _____ | $\text{Na}(\text{g}) + \text{Cl}(\text{g}) + \text{e}^- (+725 \text{ kJ})$               |
| _____ | $\text{Na}(\text{g}) + \text{e}^- + \frac{1}{2} \text{Cl}_2(\text{g}) (+603 \text{ kJ})$ |
| _____ | $\text{Na}(\text{g}) + \text{Cl}(\text{g}) (+376 \text{ kJ})$                            |
| _____ | $\text{Na}(\text{g}) + \frac{1}{2} \text{Cl}_2(\text{g}) (+107 \text{ kJ})$              |
| _____ | $\text{Na}(\text{s}) + \frac{1}{2} \text{Cl}_2(\text{g}) (+0 \text{ kJ})$                |
| _____ | $\text{NaCl}(\text{s}) (-411 \text{ kJ})$                                                |

(a) Explain in terms of ionic charge and ionic radius why MgCl<sub>2</sub> has a significantly more exothermic lattice energy than NaCl. [4]

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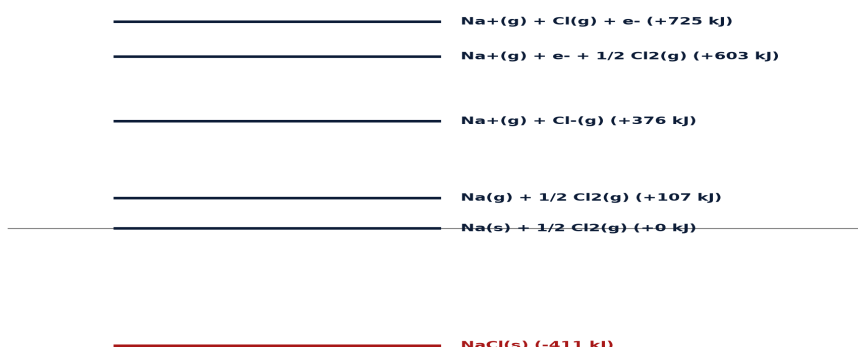
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### Q6. 6. Experimental vs Theoretical Lattice Energies: Polarization & Covalency

[5 Marks]  
WCH14/01/Jun21/Q9

For AgI: Experimental Born-Haber lattice energy =  $-889 \text{ kJ mol}^{-1}$ ; Theoretical ionic model lattice energy =  $-758 \text{ kJ mol}^{-1}$ .

#### Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Explain why the experimental lattice energy is significantly more exothermic than the theoretical value. [4]

(b) State which ion acts as the polarizing cation and which as the polarizable anion. [1]

### Q7. 7. Fajans' Rules for Ionic Polarization

[4 Marks]  
WCH14/01/Jan21/Q10

Fajans' rules summarize factors that favour covalent character in ionic compounds.

(a) State the two properties of a cation and two properties of an anion that maximize polarization. [4]

### Q8. 8. Enthalpy Change of Solution & Hydration Cycle

[5 Marks]  
WCH14/01/Oct20/Q8

For  $\text{MgSO}_4$ :  $\Delta_{\text{LE}} H = -2870 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{Mg}^{2+}) = -1920 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{SO}_4^{2-}) = -1041 \text{ kJ mol}^{-1}$ .

(a) Construct an enthalpy cycle relating  $\Delta_{\text{sol}} H$ ,  $\Delta_{\text{LE}} H$ , and  $\Delta_{\text{hyd}} H$ . [2]

(b) Calculate  $\Delta_{\text{sol}} H$  for  $\text{MgSO}_4$ . [3]

### Q9. 9. Solubility Trend of Group 2 Sulfates

[5 Marks]  
WCH14/01/Jun20/Q10

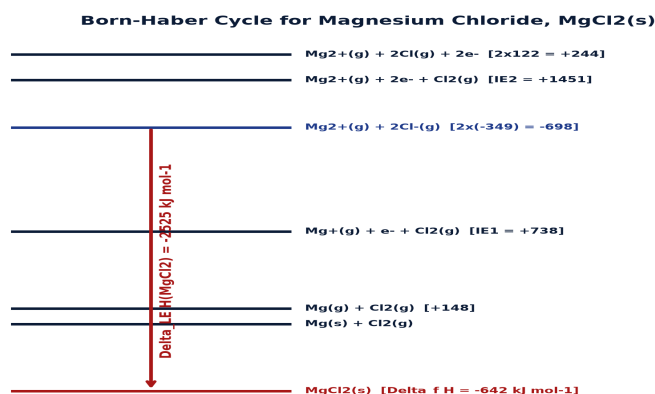
Solubility of Group 2 sulfates decreases down the group ( $\text{MgSO}_4$  soluble  $\rightarrow$   $\text{BaSO}_4$  insoluble).

(a) Explain this trend in terms of changes in lattice energy and hydration enthalpy down the group. [5]

### Q10. 10. Born-Haber Cycle Calculation for $\text{MgCl}_2$

[5 Marks]  
WCH14/01/Jan20/Q10

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_f H(\text{MgCl}_2) = -642$ ,  $\Delta_{\text{at}} H(\text{Mg}) = +148$ ,  $\text{IE}_1(\text{Mg}) = +738$ ,  $\text{IE}_2(\text{Mg}) = +1451$ ,  $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +244$ ,  $2 \times \text{EA}_1(\text{Cl}) = -698$ .



(a) Calculate the lattice energy  $\Delta_{\text{LE}} H$  for  $\text{MgCl}_2(\text{s})$ . [4]

(b) Compare  $\Delta_{\text{LE}} H(\text{MgCl}_2)$  with  $\Delta_{\text{LE}} H(\text{NaCl})$  ( $-787 \text{ kJ mol}^{-1}$ ). [1]

**Q11. 11. Solubility Trend of Group 2 Hydroxides****[ 5 Marks ]**  
**WCH14/01/Oct19/Q11**

Solubility of Group 2 hydroxides increases down the group ( $\text{Mg}(\text{OH})_2$  insoluble  $\rightarrow$   $\text{Ba}(\text{OH})_2$  soluble).

(a) Explain why the solubility trend of hydroxides is opposite to that of sulfates. [5]

**Q12. 12. Enthalpy of Hydration Definition & Trend****[ 4 Marks ]**  
**WCH14/01/Jun19/Q12**

Delta\_hyd  $H^\circ$  values ( $\text{kJ mol}^{-1}$ ):  $\text{Na}^+ = -406$ ,  $\text{Mg}^{2+} = -1920$ ,  $\text{Al}^{3+} = -4665$ .

(a) Define standard enthalpy change of hydration. [2]

(b) Explain the trend in Delta\_hyd  $H^\circ$  across  $\text{Na}^+$ ,  $\text{Mg}^{2+}$ , and  $\text{Al}^{3+}$ . [2]

**Q13. 13. Enthalpy of Atomisation Definition****[ 3 Marks ]**  
**WCH14/01/Jan19/Q13**

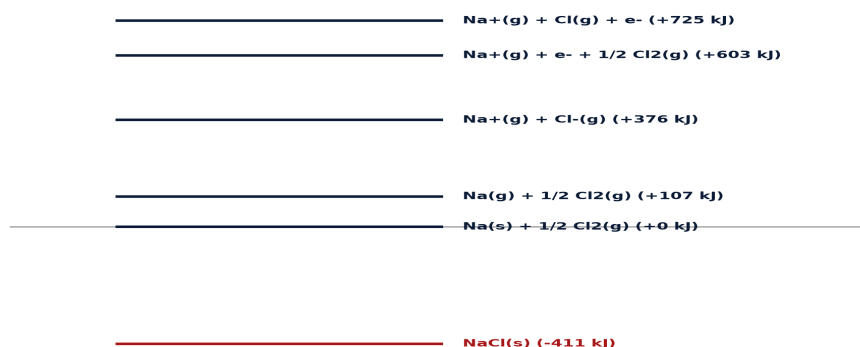
For chlorine:  $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$  (Delta\_at  $H = +122 \text{ kJ mol}^{-1}$ ).

(a) Define standard enthalpy change of atomisation. [2]

(b) Relate Delta\_at  $H$  of chlorine to its bond dissociation enthalpy. [1]

**Q14. 14. Theoretical Lattice Energy Calculation via Born-Landé Equation****[ 4 Marks ]**  
**WCH14/01/sample/Q14**

The ionic model assumes ions are perfectly spherical with purely electrostatic attraction.

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) State the two assumptions of the purely ionic model. [2]

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(b) Explain why experimental lattice energy matches theoretical value for NaCl but not for AgCl. [2]

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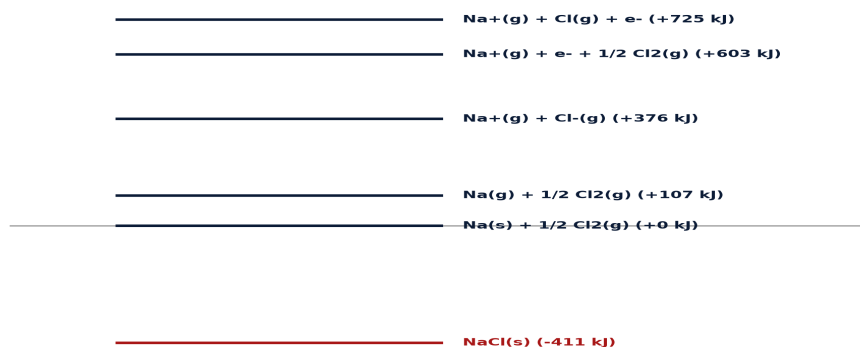


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**Q15. 15. Born-Haber Cycle for Calcium Fluoride CaF<sub>2</sub>**

[5 Marks]  
WCH14/01/Sample/Q15

Data (kJ mol<sup>-1</sup>):  $\Delta_f H(\text{CaF}_2) = -1220$ ,  $\Delta_{\text{at}} H(\text{Ca}) = +178$ ,  $\text{IE}_1(\text{Ca}) = +590$ ,  $\text{IE}_2(\text{Ca}) = +1145$ ,  $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{F}_2) = +158$ ,  $2 \times \text{EA}_1(\text{F}) = -656$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) Calculate  $\Delta_{\text{LE}} H$  for CaF<sub>2</sub>. [4]

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(b) State why 2 moles of F atoms undergo electron affinity. [1]

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[4 Marks]

**Q16. 16. Enthalpy of Solution of Sodium Hydroxide**

WCH14/01/Sample/Q16

$\text{NaOH(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$ :  $\Delta_{\text{LE}} H = -900 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{Na}^+) = -406 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{OH}^-) = -539 \text{ kJ mol}^{-1}$ .

(a) Calculate  $\Delta_{\text{sol}} H$  for NaOH. [3]

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(b) Explain why dissolving NaOH in water causes a temperature rise. [1]

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[5 Marks]

**Q17. 17. Born-Haber Cycle for Potassium Oxide K<sub>2</sub>O**

WCH14/01/Sample/Q17

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_{\text{f}} H(\text{K}_2\text{O}) = -363$ ,  $2 \times \Delta_{\text{at}} H(\text{K}) = +178$ ,  $2 \times \text{IE1}(\text{K}) = +836$ ,  $1/2 \Delta_{\text{at}} H(\text{O}_2) = +249$ ,  $\text{EA1}(\text{O}) = -141$ ,  $\text{EA2}(\text{O}) = +798$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}(\text{g}) + \text{e}^- (+725 \text{ kJ})$       |
| _____ | $\text{Na}^+(\text{g}) + \text{e}^- + 1/2 \text{Cl}_2(\text{g}) (+603 \text{ kJ})$ |
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) (+376 \text{ kJ})$                  |
| _____ | $\text{Na}(\text{g}) + 1/2 \text{Cl}_2(\text{g}) (+107 \text{ kJ})$                |
| _____ | $\text{Na}(\text{s}) + 1/2 \text{Cl}_2(\text{g}) (+0 \text{ kJ})$                  |
| _____ | $\text{NaCl(s)} (-411 \text{ kJ})$                                                 |

(a) Calculate  $\Delta_{\text{LE}} H$  for  $\text{K}_2\text{O(s)}$ . [5]

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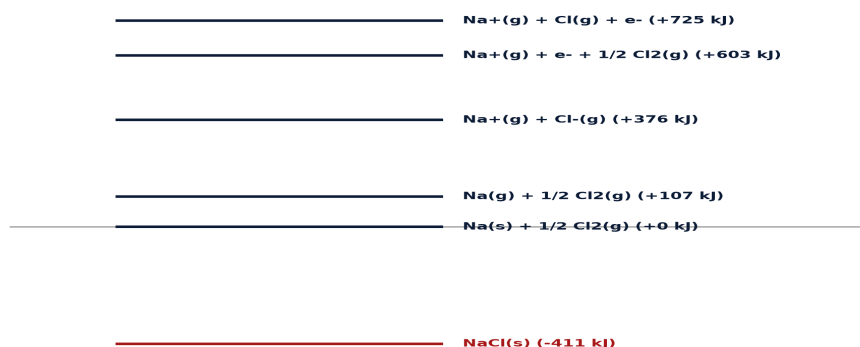
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[3 Marks]

**Q18. 18. Polarizing Power Comparison: Li<sup>+</sup> vs K<sup>+</sup>**

WCH14/01/Sample/Q18

LiI has experimental lattice energy  $-849 \text{ kJ mol}^{-1}$  (theoretical  $-738$ ), whereas KI is  $-649$  (theoretical  $-637$ ).

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) Explain why LiI has much greater covalent character than KI. [3]

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**Q19. 19. Enthalpy of Hydration of Halide Ions****[4 Marks]****WCH14/01/Sample/Q19**

Delta\_hyd H° (kJ mol<sup>-1</sup>): F<sup>-</sup> = -506, Cl<sup>-</sup> = -364, Br<sup>-</sup> = -335, I<sup>-</sup> = -293.

(a) Explain the trend in hydration enthalpy down Group 17. [4]

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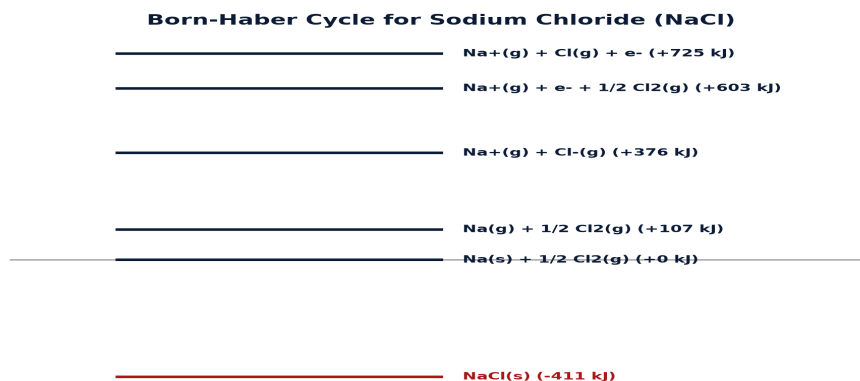
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**Q20. 20. Born-Haber Cycle for Aluminum Oxide Al<sub>2</sub>O<sub>3</sub>****[5 Marks]****WCH14/01/Sample/Q20**

Al<sub>2</sub>O<sub>3</sub> has an exceptionally large exothermic lattice energy (Delta\_LE H = -15116 kJ mol<sup>-1</sup>).



(a) Explain why Al<sub>2</sub>O<sub>3</sub> has a vastly more exothermic lattice energy than NaCl (-787 kJ mol<sup>-1</sup>). [5]

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[4 Marks]

**Q21. 21. Thermal Decomposition of Group 2 Nitrates**

WCH14/01/Sample/Q21

$2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ . Decomposition temperature increases down Group 2.

(a) Explain why  $\text{Mg}(\text{NO}_3)_2$  decomposes at a lower temperature than  $\text{Ba}(\text{NO}_3)_2$ . [4]

[4 Marks]

**Q22. 22. Solubility of Calcium Hydroxide vs Calcium Sulfate**

WCH14/01/Sample/Q22

$\text{Ca}(\text{OH})_2$  is sparingly soluble (pH ~12), while  $\text{CaSO}_4$  is poorly soluble.

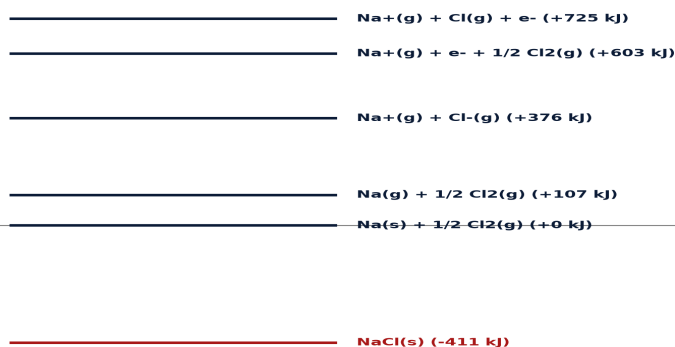
(a) Relate their lattice energies and hydration enthalpies to explain why  $\text{Ca}(\text{OH})_2$  is more soluble than  $\text{CaSO}_4$ . [4]

[5 Marks]

**Q23. 23. Born-Haber Cycle for Lithium Iodide LiI**

WCH14/01/Sample/Q23

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_f H(\text{LiI}) = -270$ ,  $\Delta_{\text{at}} H(\text{Li}) = +159$ ,  $\text{IE}(\text{Li}) = +520$ ,  $\frac{1}{2} \Delta_{\text{at}} H(\text{I}_2) = +107$ ,  $\text{EA}(\text{I}) = -295$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) Calculate  $\Delta_{\text{LE}} H$  for  $\text{LiI}(\text{s})$ . [4]

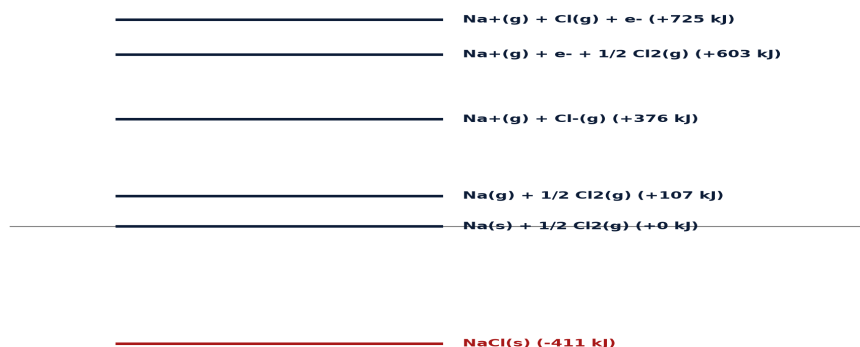
(b) Explain why LiI is soluble in organic solvents like ethanol. [1]

#### Q24. 24. Enthalpy of Hydration vs Lattice Energy for BaCl<sub>2</sub>

[4 Marks]  
WCH14/01/Sample/Q24

$\text{BaCl}_2(\text{s}) \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$ :  $\Delta_{\text{LE}} H = -2056 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{Ba}^{2+}) = -1305 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{Cl}^{-}) = -364 \text{ kJ mol}^{-1}$ .

##### Born-Haber Cycle for Sodium Chloride (NaCl)



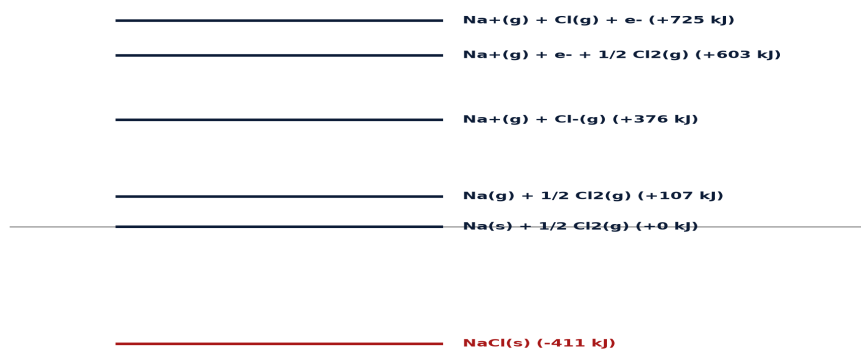
(a) Calculate  $\Delta_{\text{sol}} H$  for BaCl<sub>2</sub>. [3]

(b) State whether dissolution is exothermic or endothermic. [1]

#### Q25. 25. Born-Haber Cycle for Copper(II) Oxide CuO

[5 Marks]  
WCH14/01/Sample/Q25

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_{\text{f}} H(\text{CuO}) = -157$ ,  $\Delta_{\text{at}} H(\text{Cu}) = +338$ ,  $\text{IE}_1(\text{Cu}) = +746$ ,  $\text{IE}_2(\text{Cu}) = +1958$ ,  $1/2 \Delta_{\text{at}} H(\text{O}_2) = +249$ ,  $\text{EA}_1(\text{O}) + \text{EA}_2(\text{O}) = +657$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) Calculate  $\Delta_{\text{LEH}}$  for  $\text{CuO}(\text{s})$ . [5]

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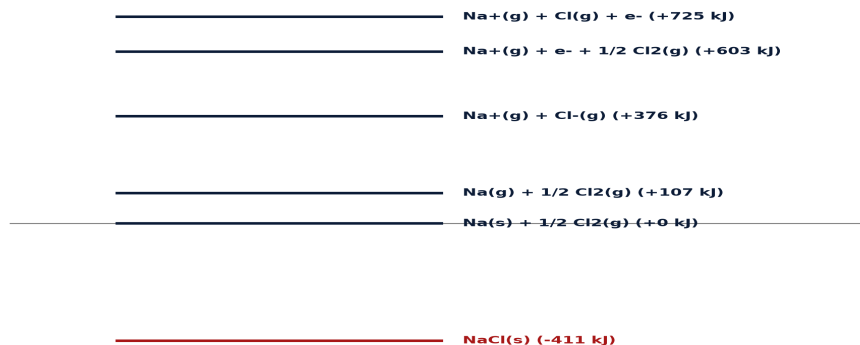
## SECTION B: TIER 2 A\* CHALLENGE QUESTIONS (HARD &amp; EXTENDED) — Q26 to Q50

## Q26. 26. A\* Challenge: Born-Haber Cycle with Unknown Electron Affinity

[5 Marks]  
WCH14/01/Hard/Q26

For metal bromide  $MBr_2$ :  $\Delta_f H = -524 \text{ kJ mol}^{-1}$ ,  $\Delta_{at} H(M) = +178$ ,  $IE_1(M) = +590$ ,  $IE_2(M) = +1145$ ,  $\Delta_{at} H(Br_2) = +112$ ,  $\Delta_{LE} H = -2150$ .

## Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Calculate the first electron affinity of bromine,  $EA_1(Br)$ , in  $\text{kJ mol}^{-1}$ . [5]

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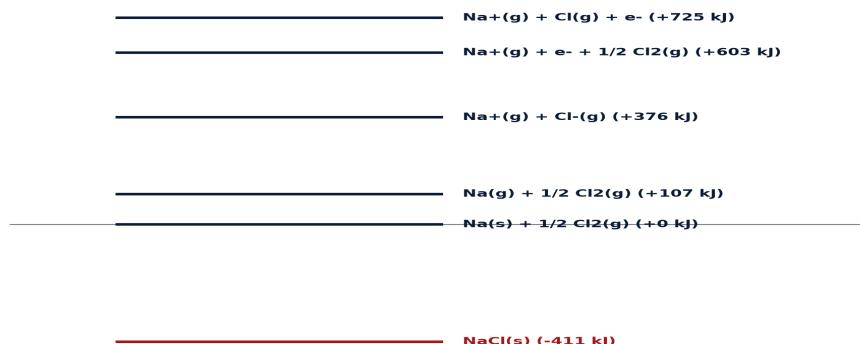
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## Q27. 27. A\* Challenge: Percent Covalent Character from Lattice Energy Comparison

[5 Marks]  
WCH14/01/Hard/Q27

For  $AgBr$ : Experimental Born-Haber lattice energy =  $-905 \text{ kJ mol}^{-1}$ ; Theoretical ionic model lattice energy =  $-758 \text{ kJ mol}^{-1}$ .

## Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Calculate the percentage difference between experimental and theoretical lattice energy. [2]

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(b) Explain the structural origin of this discrepancy in detail. [3]

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**Q28. 28. A\* Challenge: Enthalpy of Solution Temperature Dependence & Entropy**

[5 Marks]  
WCH14/01/Hard/Q28

For KI:  $\Delta_{\text{LE}} H = -649 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{K}^+) = -322$ ,  $\Delta_{\text{hyd}} H(\text{I}^-) = -293$ .

(a) Calculate  $\Delta_{\text{sol}} H$  for KI. [2]

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(b) Predict how the solubility of KI changes with increasing temperature using Le Chatelier and entropy. [3]

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**Q29. 29. A\* Challenge: Lattice Energy of Hypothetical NaCl<sub>2</sub>**

[6 Marks]  
WCH14/01/Hard/Q29

Consider why NaCl<sub>2</sub>(s) does not form despite its high theoretical lattice energy.

**Born-Haber Cycle for Sodium Chloride (NaCl)**

|       |                                                                                   |
|-------|-----------------------------------------------------------------------------------|
| _____ | $\text{Na}(\text{g}) + \text{Cl}(\text{g}) + \text{e}^- (+725 \text{ kJ})$        |
| _____ | $\text{Na}(\text{g}) + \text{e}^- + 1/2 \text{ Cl}_2(\text{g}) (+603 \text{ kJ})$ |
| _____ | $\text{Na}(\text{g}) + \text{Cl}(\text{g}) (+376 \text{ kJ})$                     |
| _____ | $\text{Na}(\text{g}) + 1/2 \text{ Cl}_2(\text{g}) (+107 \text{ kJ})$              |
| _____ | $\text{Na}(\text{s}) + 1/2 \text{ Cl}_2(\text{g}) (+0 \text{ kJ})$                |
| _____ | $\text{NaCl}(\text{s}) (-411 \text{ kJ})$                                         |

(a) Estimate  $\Delta_{\text{LE}} H$  for NaCl<sub>2</sub>(s) assuming it is similar to MgCl<sub>2</sub> (−2500 kJ mol<sup>−1</sup>). [1]

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(b) Explain why the second ionisation energy of sodium ( $\text{IE}_2 = +4562 \text{ kJ mol}^{-1}$ ) makes  $\Delta_{\text{f}} H(\text{NaCl}_2)$  highly endothermic. [3]

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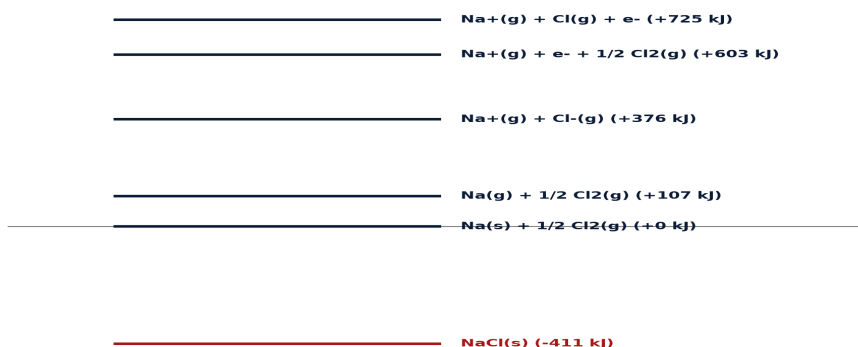
(c) Conclude why NaCl(s) forms instead of NaCl<sub>2</sub>(s). [2]

**Q30. 30. A\* Challenge: Hydration Enthalpy vs Lattice Energy  
Competition in Halides**

**[5 Marks]**  
**WCH14/01/Hard/Q30**

Enthalpies of solution  $\Delta_{\text{sol}} H$  ( $\text{kJ mol}^{-1}$ ):  $\text{LiF} = +4.7$ ,  $\text{LiCl} = -37.0$ ,  $\text{LiBr} = -48.8$ ,  $\text{LiI} = -63.3$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**



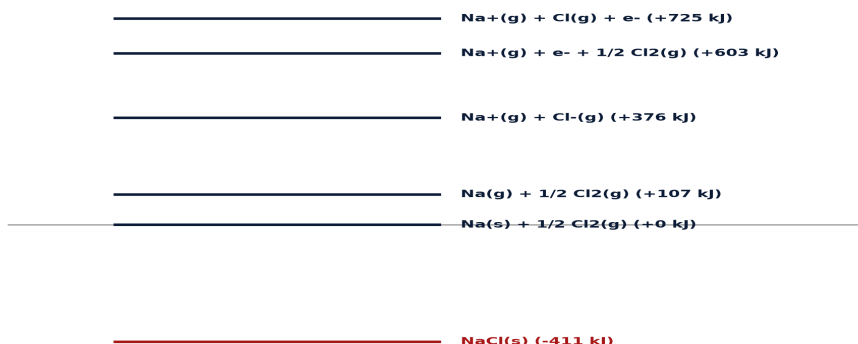
(a) Explain the trend in  $\Delta_{\text{sol}} H$  down the lithium halides. [5]

**Q31. 31. A\* Challenge: Born-Haber Cycle for Calcium Oxide  
CaO**

**[5 Marks]**  
**WCH14/01/Hard/Q31**

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_f H(\text{CaO}) = -635$ ,  $\Delta_{\text{at}} H(\text{Ca}) = +178$ ,  $\text{IE}_1(\text{Ca}) = +590$ ,  $\text{IE}_2(\text{Ca}) = +1145$ ,  $\frac{1}{2} \Delta_{\text{at}} H(\text{O}_2) = +249$ ,  $\text{EA}_1(\text{O}) = -141$ ,  $\text{EA}_2(\text{O}) = +798$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**



(a) Calculate  $\Delta_{\text{LE}} H$  for  $\text{CaO}(\text{s})$ . [4]

(b) Compare  $\Delta_{\text{LE}} H(\text{CaO})$  with  $\Delta_{\text{LE}} H(\text{NaCl})$  ( $-787 \text{ kJ mol}^{-1}$ ). [1]

**Q32. 32. A\* Challenge: Polarization Effect on Thermal Stability of Carbonates**

[5 Marks]  
WCH14/01/Hard/Q32

Compare thermal decomposition of  $\text{Na}_2\text{CO}_3$  (does not decompose at  $800^\circ\text{C}$ ) vs  $\text{CaCO}_3$  (decomposes at  $840^\circ\text{C}$ ) vs  $\text{MgCO}_3$  (decomposes at  $300^\circ\text{C}$ ).

(a) Explain why  $\text{Na}^+$  causes less polarization than  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ . [3]

(b) Explain why  $\text{MgCO}_3$  decomposes at a much lower temperature than  $\text{CaCO}_3$ . [2]

**Q33. 33. A\* Challenge: Born-Haber Cycle for Barium Chloride  $\text{BaCl}_2$**

[5 Marks]  
WCH14/01/Hard/Q33

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_{\text{f}} H(\text{BaCl}_2) = -858$ ,  $\Delta_{\text{at}} H(\text{Ba}) = +180$ ,  $\text{IE}_1(\text{Ba}) = +503$ ,  $\text{IE}_2(\text{Ba}) = +965$ ,  $2 \times \frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +244$ ,  $2 \times \text{EA}_1(\text{Cl}) = -698$ .

**Born-Haber Cycle for Sodium Chloride ( $\text{NaCl}$ )**

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}(\text{g}) + \text{e}^-$ (+725 kJ)               |
| _____ | $\text{Na}^+(\text{g}) + \text{e}^- + \frac{1}{2} \text{Cl}_2(\text{g})$ (+603 kJ) |
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$ (+376 kJ)                          |
| _____ | $\text{Na}(\text{g}) + \frac{1}{2} \text{Cl}_2(\text{g})$ (+107 kJ)                |
| _____ | $\text{Na}(\text{s}) + \frac{1}{2} \text{Cl}_2(\text{g})$ (+0 kJ)                  |
| _____ | $\text{NaCl}(\text{s})$ (-411 kJ)                                                  |

(a) Calculate  $\Delta_{\text{LE}} H$  for  $\text{BaCl}_2(\text{s})$ . [4]

(b) Compare  $\Delta_{\text{LE}} H(\text{BaCl}_2)$  with  $\Delta_{\text{LE}} H(\text{MgCl}_2)$  ( $-2525 \text{ kJ mol}^{-1}$ ). [1]

**Q34. 34. A\* Challenge: Hydration Enthalpy vs Ion Size in Transition Metal Ions****[5 Marks]****WCH14/01/Hard/Q34**

Delta\_hyd H° (kJ mol<sup>-1</sup>): Fe<sup>2+</sup> = -1946, Fe<sup>3+</sup> = -4430.

- (a) Explain why Fe<sup>3+</sup> has a vastly more exothermic hydration enthalpy than Fe<sup>2+</sup>. [3]

- (b) Predict how this affects the acidity of [Fe(H<sub>2</sub>O)<sub>6</sub>]<sup>3+</sup> vs [Fe(H<sub>2</sub>O)<sub>6</sub>]<sup>2+</sup> solutions. [2]

**Q35. 35. A\* Challenge: Enthalpy Cycle for Anhydrous vs Hydrated Copper Sulfate****[5 Marks]****WCH14/01/Hard/Q35**

CuSO<sub>4</sub>(s) + 5H<sub>2</sub>O(l) → CuSO<sub>4</sub>·5H<sub>2</sub>O(s). Delta\_sol H(CuSO<sub>4</sub>(s)) = -66.5 kJ mol<sup>-1</sup>; Delta\_sol H(CuSO<sub>4</sub>·5H<sub>2</sub>O(s)) = +11.7 kJ mol<sup>-1</sup>.

- (a) Construct a Hess's law cycle to calculate Delta H\_hydration for CuSO<sub>4</sub>(s) → CuSO<sub>4</sub>·5H<sub>2</sub>O(s). [4]

- (b) Why cannot Delta H\_hydration be measured directly by calorimetry? [1]

**Q36. 36. A\* Challenge: Enthalpy of Solution of Potassium Nitrate KNO<sub>3</sub>****[5 Marks]****WCH14/01/Hard/Q36**

KNO<sub>3</sub>(s) → K<sup>+</sup>(aq) + NO<sub>3</sub><sup>-</sup>(aq): Delta\_LE H = -687 kJ mol<sup>-1</sup>, Delta\_hyd H(K<sup>+</sup>) = -322, Delta\_hyd H(NO<sub>3</sub><sup>-</sup>) = -314.

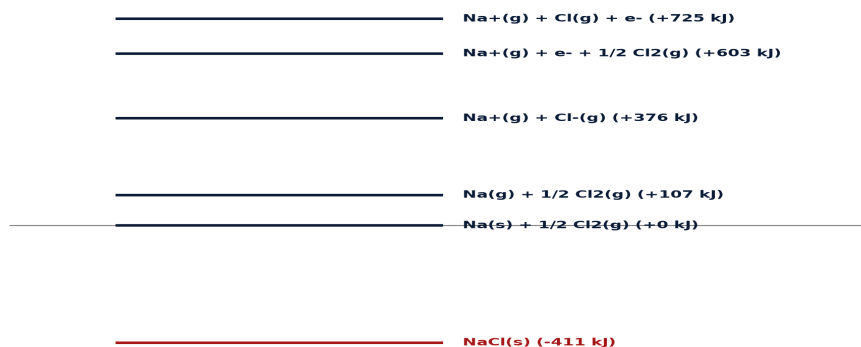
- (a) Calculate Delta\_sol H for KNO<sub>3</sub>. [2]

- (b) If Delta S\_system = +115 J K<sup>-1</sup> mol<sup>-1</sup>, calculate Delta S\_total at 298 K and explain why KNO<sub>3</sub> dissolves. [3]

**Q37. 37. A\* Challenge: Born-Haber Cycle for Silver Chloride  
AgCl**

[5 Marks]  
WCH14/01/Hard/Q37

Data (kJ mol<sup>-1</sup>):  $\Delta_f H(\text{AgCl}) = -127$ ,  $\Delta_{\text{at}} H(\text{Ag}) = +285$ ,  $IE_1(\text{Ag}) = +731$ ,  $\frac{1}{2} \Delta_{\text{at}} H(\text{Cl}_2) = +122$ ,  $EA_1(\text{Cl}) = -349$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**


(a) Calculate experimental Born-Haber  $\Delta_{\text{LE}} H$  for AgCl. [3]

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(b) The theoretical ionic lattice energy for AgCl is -770 kJ mol<sup>-1</sup>. Account for the difference. [2]

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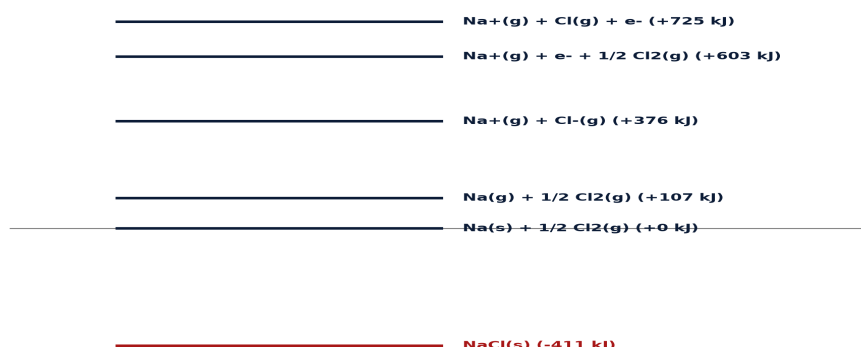
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**Q38. 38. A\* Challenge: Enthalpy of Atomisation of Metals vs  
Lattice Energy**

[5 Marks]  
WCH14/01/Hard/Q38

$\Delta_{\text{at}} H$  (kJ mol<sup>-1</sup>): Na = +107, Mg = +148, Al = +326.

**Born-Haber Cycle for Sodium Chloride (NaCl)**


(a) Explain the trend in enthalpy of atomisation across Period 3 metals. [3]

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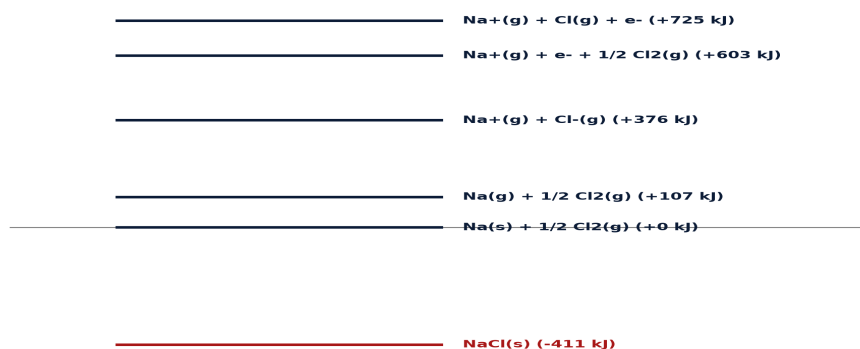
(b) Relate metallic bond strength to charge density of metal cations and delocalised electron count. [2]

**Q39. 39. A\* Challenge: Born-Haber Cycle for Zinc Sulfide ZnS**

[5 Marks]  
WCH14/01/Hard/Q39

Data (kJ mol<sup>-1</sup>):  $\Delta_f H(\text{ZnS}) = -206$ ,  $\Delta_{\text{at}} H(\text{Zn}) = +131$ ,  $\text{IE}_1(\text{Zn}) + \text{IE}_2(\text{Zn}) = +2735$ ,  $\Delta_{\text{at}} H(\text{S}) = +279$ ,  $\text{EA}_1(\text{S}) + \text{EA}_2(\text{S}) = +332$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**



(a) Calculate  $\Delta_{\text{LE}} H$  for  $\text{ZnS}(\text{s})$ . [4]

(b) Explain why ZnS adopts a covalent zinc blende structure rather than a purely ionic NaCl lattice. [1]

**Q40. 40. A\* Challenge: Dissolution Energetics of Calcium Chloride CaCl<sub>2</sub>**

[5 Marks]  
WCH14/01/Hard/Q40

$\text{CaCl}_2(\text{s}) \rightarrow \text{Ca}^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$ :  $\Delta_{\text{LE}} H = -2258 \text{ kJ mol}^{-1}$ ,  $\Delta_{\text{hyd}} H(\text{Ca}^{2+}) = -1577$ ,  $\Delta_{\text{hyd}} H(\text{Cl}^{-}) = -364$ .

(a) Calculate  $\Delta_{\text{sol}} H$  for  $\text{CaCl}_2$ . [3]

(b) Why is  $\text{CaCl}_2$  used in self-heating food cans and de-icing roads? [2]

**Q41. 41. A\* Challenge: Polarization and Thermal Decomposition of Hydroxides**
**[5 Marks]**  
**WCH14/01/Hard/Q41**

$\text{Mg}(\text{OH})_2(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2\text{O}(\text{g})$  occurs at 350 °C, whereas  $\text{Ba}(\text{OH})_2$  melts at 780 °C without decomposing.

(a) Explain in terms of polarizing power of  $\text{Mg}^{2+}$  vs  $\text{Ba}^{2+}$  why  $\text{Mg}(\text{OH})_2$  decomposes easily. [5]

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**Q42. 42. A\* Challenge: Enthalpy of Hydration of Gaseous Protons  $\text{H}^+(\text{g})$** 
**[5 Marks]**  
**WCH14/01/Hard/Q42**

The hydration enthalpy of  $\text{H}^+(\text{g})$  is exceptionally exothermic:  $\Delta_{\text{hyd}} \text{H}(\text{H}^+) = -1090 \text{ kJ mol}^{-1}$ .

(a) Explain why  $\text{H}^+(\text{g})$  has a vastly more exothermic hydration enthalpy than  $\text{Na}^+(\text{g})$  ( $-406 \text{ kJ mol}^{-1}$ ). [3]

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(b) Write the chemical equation for  $\text{H}^+(\text{g})$  hydration forming  $\text{H}_3\text{O}^+(\text{aq})$ . [2]

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**Q43. 43. A\* Challenge: Born-Haber Cycle for Sodium Fluoride NaF**
**[5 Marks]**  
**WCH14/01/Hard/Q43**

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_{\text{f}} \text{H}(\text{NaF}) = -574$ ,  $\Delta_{\text{at}} \text{H}(\text{Na}) = +107$ ,  $\text{IE1}(\text{Na}) = +496$ ,  $1/2 \Delta_{\text{at}} \text{H}(\text{F}_2) = +79$ ,  $\text{EA1}(\text{F}) = -328$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

|       |                                                                                    |
|-------|------------------------------------------------------------------------------------|
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}(\text{g}) + \text{e}^- (+725 \text{ kJ})$       |
| _____ | $\text{Na}^+(\text{g}) + \text{e}^- + 1/2 \text{Cl}_2(\text{g}) (+603 \text{ kJ})$ |
| _____ | $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) (+376 \text{ kJ})$                  |
| _____ | $\text{Na}(\text{g}) + 1/2 \text{Cl}_2(\text{g}) (+107 \text{ kJ})$                |
| _____ | $\text{Na}(\text{s}) + 1/2 \text{Cl}_2(\text{g}) (+0 \text{ kJ})$                  |
| _____ | $\text{NaCl}(\text{s}) (-411 \text{ kJ})$                                          |

(a) Calculate  $\Delta_{\text{LE}} \text{H}$  for  $\text{NaF}(\text{s})$ . [4]

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(b) Compare  $\Delta_{\text{LE}} H(\text{NaF})$  with  $\Delta_{\text{LE}} H(\text{NaCl})$  ( $-787 \text{ kJ mol}^{-1}$ ). [1]

**Q44. 44. A\* Challenge: Solvation Enthalpies in Non-Aqueous Solvents**

[5 Marks]  
WCH14/01/Hard/Q44

$\Delta_{\text{sol}} H(\text{LiCl})$  in liquid ammonia is  $-120 \text{ kJ mol}^{-1}$  compared to  $-37 \text{ kJ mol}^{-1}$  in water.

(a) Explain why solvation enthalpy of  $\text{Li}^+$  is more exothermic in ammonia than in water. [3]

(b) Relate nitrogen lone pair electron-donating ability to oxygen in water. [2]

**Q45. 45. A\* Challenge: Theoretical vs Experimental Lattice Energies of Transition Metal Halides**

[5 Marks]  
WCH14/01/Hard/Q45

$\Delta_{\text{LE}} H$  ( $\text{kJ mol}^{-1}$ ):  $\text{MnCl}_2$  (exp  $-2520$ , theo  $-2480$ );  $\text{FeCl}_2$  (exp  $-2530$ , theo  $-2490$ );  $\text{CuCl}_2$  (exp  $-2750$ , theo  $-2580$ ).

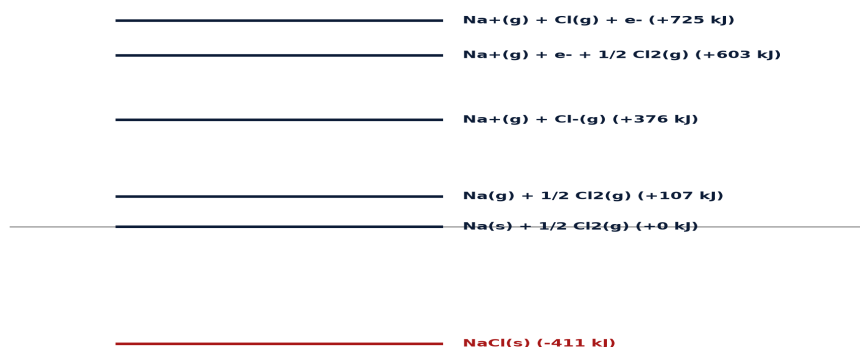
(a) Explain why  $\text{CuCl}_2$  shows a much larger discrepancy ( $170 \text{ kJ mol}^{-1}$ ) than  $\text{MnCl}_2$  ( $40 \text{ kJ mol}^{-1}$ ). [3]

(b) Relate this to Crystal Field Stabilization Energy (CFSE) and Jahn-Teller distortion. [2]

**Q46. 46. A\* Challenge: Born-Haber Cycle for Lead(II) Chloride  $\text{PbCl}_2$**

[5 Marks]  
WCH14/01/Hard/Q46

Data ( $\text{kJ mol}^{-1}$ ):  $\Delta_f H(\text{PbCl}_2) = -359$ ,  $\Delta_{\text{at}} H(\text{Pb}) = +195$ ,  $\text{IE}_1(\text{Pb}) + \text{IE}_2(\text{Pb}) = +2190$ ,  $\Delta_{\text{at}} H(\text{Cl}_2) = +244$ ,  $2 \times \text{EA}_1(\text{Cl}) = -698$ .

**Born-Haber Cycle for Sodium Chloride (NaCl)**

(a) Calculate experimental  $\Delta_{\text{LEH}}$  for  $\text{PbCl}_2(\text{s})$ . [4]

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(b) Why is  $\text{PbCl}_2$  insoluble in cold water? [1]

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**Q47. 47. A\* Challenge: Calculating Enthalpy of Solution from Calorimetry Data**

[5 Marks]  
**WCH14/01/Hard/Q47**

5.00 g of anhydrous  $\text{CaCl}_2$  ( $M_r = 111.0$ ) was dissolved in 100 cm<sup>3</sup> of water in a calorimeter. Temperature rose by 8.4 °C.

(Specific heat capacity  $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$ )

(a) Calculate  $q = m c \Delta T$  in Joules. [2]

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(b) Calculate moles of  $\text{CaCl}_2$  and deduce  $\Delta_{\text{solH}}$  in kJ mol<sup>-1</sup>. [3]

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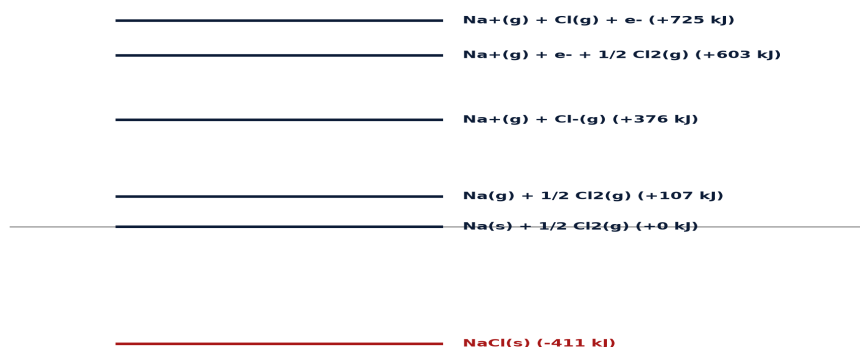
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**Q48. 48. A\* Challenge: Born-Haber Cycle for Potassium Iodide KI**

[5 Marks]  
**WCH14/01/Hard/Q48**

Data (kJ mol<sup>-1</sup>):  $\Delta_f H(\text{KI}) = -328$ ,  $\Delta_{\text{at}} H(\text{K}) = +89$ ,  $\text{IE}(\text{K}) = +418$ ,  $1/2 \Delta_{\text{at}} H(\text{I}_2) = +107$ ,  $\text{EA}(\text{I}) = -295$ .

### Born-Haber Cycle for Sodium Chloride (NaCl)



(a) Calculate  $\Delta_{\text{LEH}}$  for  $\text{KI(s)}$ . [4]

(b) Why does KI have a theoretical lattice energy almost identical to experimental? [1]

### Q49. 49. A \* Challenge: Hydration Enthalpy vs Ionic Size Grid Analysis

**[5 Marks]**

WCH14/01/Hard/Q49

Predict and rank  $\Delta_{\text{hyd}} H^\circ$  for  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Mg}^{2+}$ ,  $\text{Ca}^{2+}$ .

(a) Rank the four ions in order of increasingly exothermic hydration enthalpy with full justification. [5]

### Q50. 50. A\* Challenge: Complete Energetics Master Synthesis

**[ 6 Marks ]**

WCH14/01/Hard/Q50

A student investigates anhydrous  $\text{MgF}_2(\text{s})$ .

Data (kJ mol<sup>-1</sup>): Delta\_f H(MgF2) = -1124, Delta\_at H(Mg) = +148, IE1(Mg)+IE2(Mg) = +2189, Delta\_at H(F2) = +158, EA1(F) = -328.

(a) Calculate  $\Delta_{LEH}$  for  $MgF_2(s)$ . [3]

(b) Given  $\Delta_{\text{hyd}} H(\text{Mg}^{2+}) = -1920$  and  $\Delta_{\text{hyd}} H(\text{F}^{-}) = -506 \text{ kJ mol}^{-1}$ , calculate  $\Delta_{\text{sol}} H$  for  $\text{MgF}_2$ . [3]

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## SECTION C: WORKED MARK SCHEME (Q1–Q50) — 12B

| Q   | MARKING SCHEME & ACCEPTABLE ANSWERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MARKS |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Q1  | Q1: 1. Lattice Energy Definition<br>1. (a) Enthalpy change when one mole of an ionic solid is formed from its gaseous ions under standard conditions (2).<br>1. (b) Bond formation between oppositely charged gaseous ions is an exothermic process (1).                                                                                                                                                                                                                                                               | [3]   |
| Q2  | Q2: 2. Born-Haber Cycle Calculation for NaCl<br>2. (a) $\Delta_f H = \Delta_{at} H(\text{Na}) + IE1(\text{Na}) + 1/2 \Delta_{at} H(\text{Cl}_2) + EA1(\text{Cl}) + \Delta_{LE} H$ (1).<br>$-411 = +107 + 496 + 122 - 349 + \Delta_{LE} H = +376 + \Delta_{LE} H$ (2). $\Delta_{LE} H = -411 - 376 = -787 \text{ kJ mol}^{-1}$ (1).<br>2. (b) More exothermic lattice energy gives higher melting point (1).                                                                                                            | [5]   |
| Q3  | Q3: 3. First Electron Affinity Definition & Sign<br>3. (a) Enthalpy change when one mole of electrons is added to one mole of gaseous atoms to form one mole of gaseous $\text{I}^-$ ions (2).<br>3. (b) Attracting incoming electron to positively charged nucleus releases energy (1).                                                                                                                                                                                                                               | [3]   |
| Q4  | Q4: 4. Second Electron Affinity Endothermic Nature<br>4. (a) $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$ (1).<br>4. (b) An electron is being added to a negatively charged $\text{O}^-$ ion; energy is required to overcome electrostatic repulsion (2).                                                                                                                                                                                                                                   | [3]   |
| Q5  | Q5: 5. Lattice Energy Comparison: NaCl vs $\text{MgCl}_2$<br>5. (a) $\text{Mg}^{2+}$ has a higher ionic charge (+2) than $\text{Na}^+$ (+1) (1). $\text{Mg}^{2+}$ has a smaller ionic radius than $\text{Na}^+$ (1). $\text{Mg}^{2+}$ has a much higher charge density (1). Greater electrostatic attraction between $\text{Mg}^{2+}$ and $\text{Cl}^-$ ions (1).                                                                                                                                                      | [4]   |
| Q6  | Q6: 6. Experimental vs Theoretical Lattice Energies: Polarization & Covalency<br>6. (a) $\text{Ag}^+$ has high charge density and polarizes the large, highly polarizable $\text{I}^-$ electron cloud (2). This distorts the electron cloud, introducing significant covalent character (1). Covalent bonds provide additional bonding strength beyond ionic attraction (1).<br>6. (b) $\text{Ag}^+$ is polarizing cation; $\text{I}^-$ is polarizable anion (1).                                                      | [5]   |
| Q7  | Q7: 7. Fajans' Rules for Ionic Polarization<br>7. (a) Cation: Small ionic radius (1) and high ionic charge (high charge density) (1). Anion: Large ionic radius (1) and high ionic charge (1).                                                                                                                                                                                                                                                                                                                         | [4]   |
| Q8  | Q8: 8. Enthalpy Change of Solution & Hydration Cycle<br>8. (a) Enthalpy cycle: $\text{MgSO}_4(\text{s}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{SO}_4^{2-}(\text{g}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ (2).<br>8. (b) $\Delta_{\text{sol}} H = \text{Sum } \Delta_{\text{hyd}} H - \Delta_{LE} H = (-1920 - 1041) - (-2870) = -2961 - (-2870) = -91 \text{ kJ mol}^{-1}$ (3).                                                                                                  | [5]   |
| Q9  | Q9: 9. Solubility Trend of Group 2 Sulfates<br>9. (a) Cation radius increases down Group 2 ( $\text{Mg}^{2+}$ to $\text{Ba}^{2+}$ ) (1). Both lattice energy and hydration enthalpy become less exothermic down group (1). Sulfate ion $\text{SO}_4^{2-}$ is very large, so lattice energy decreases only slightly (1). Hydration enthalpy of cation decreases significantly due to larger radius/lower charge density (1). $\Delta_{\text{sol}} H$ becomes more endothermic (less negative), reducing solubility (1). | [5]   |
| Q10 | Q10: 10. Born-Haber Cycle Calculation for $\text{MgCl}_2$<br>10. (a) $\Delta_f H = \text{Sum atomisation}/IE/EA + \Delta_{LE} H$ (1). $-642 = +148 + 738 + 1451 + 244 - 698 + \Delta_{LE} H = +1883 + \Delta_{LE} H$ (2). $\Delta_{LE} H = -642 - 1883 = -2525 \text{ kJ mol}^{-1}$ (1).<br>10. (b) $\text{Mg}^{2+}$ has higher charge (+2 vs +1) and smaller radius, leading to much stronger lattice (1).                                                                                                            | [5]   |
| Q11 | Q11: 11. Solubility Trend of Group 2 Hydroxides<br>11. (a) $\text{OH}^-$ is a small anion compared to $\text{SO}_4^{2-}$ (1). Down Group 2, as cation radius increases, lattice energy decreases significantly (1). Hydration enthalpy of cation also decreases (1). Decrease in lattice energy dominates over decrease in hydration enthalpy (1). $\Delta_{\text{sol}} H$ becomes more exothermic / less endothermic down group, increasing solubility (1).                                                           | [5]   |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |
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| <b>Q12</b> | Q12: 12. Enthalpy of Hydration Definition & Trend<br>12. (a) Enthalpy change when one mole of gaseous ions is dissolved in water to form an infinitely dilute solution under standard conditions (2).<br>12. (b) Charge increases (+1 to +3) and ionic radius decreases (1), increasing charge density and ion-dipole attraction to water molecules (1).                                                                                                                                        | <b>[4]</b> |
| <b>Q13</b> | Q13: 13. Enthalpy of Atomisation Definition<br>13. (a) Enthalpy change when one mole of gaseous atoms is formed from an element in its standard state under standard conditions (2).<br>13. (b) $\Delta_{\text{at}} \text{H}(\text{Cl}_2)$ is half the bond dissociation enthalpy of $\text{Cl}-\text{Cl}$ (1).                                                                                                                                                                                 | <b>[3]</b> |
| <b>Q14</b> | Q14: 14. Theoretical Lattice Energy Calculation via Born-Landé Equation<br>14. (a) 1. Ions are 100% spherical (1). 2. Bonding is 100% ionic with zero electron cloud distortion / zero covalent character (1).<br>14. (b) $\text{Na}^+$ is non-polarizing so $\text{NaCl}$ is almost 100% ionic (1). $\text{Ag}^+$ is strongly polarizing, inducing significant covalent character in $\text{AgCl}$ (1).                                                                                        | <b>[4]</b> |
| <b>Q15</b> | Q15: 15. Born-Haber Cycle for Calcium Fluoride $\text{CaF}_2$<br>15. (a) $-1220 = +178 + 590 + 1145 + 158 - 656 + \Delta_{\text{LE}} \text{H} = +1415 + \Delta_{\text{LE}} \text{H}$ (3). $\Delta_{\text{LE}} \text{H} = -1220 - 1415 = -2635 \text{ kJ mol}^{-1}$ (1).<br>15. (b) $\text{Ca}^{2+}$ requires 2 $\text{F}^-$ ions for charge neutrality (1).                                                                                                                                     | <b>[5]</b> |
| <b>Q16</b> | Q16: 16. Enthalpy of Solution of Sodium Hydroxide<br>16. (a) $\Delta_{\text{sol}} \text{H} = (-406 - 539) - (-900) = -945 - (-900) = -45 \text{ kJ mol}^{-1}$ (3).<br>16. (b) Reaction is exothermic ( $\Delta_{\text{sol}} \text{H} < 0$ ), releasing heat to the solution (1).                                                                                                                                                                                                                | <b>[4]</b> |
| <b>Q17</b> | Q17: 17. Born-Haber Cycle for Potassium Oxide $\text{K}_2\text{O}$<br>17. (a) Sum atomisation/IE/EA = $+178 + 836 + 249 - 141 + 798 = +1920 \text{ kJ mol}^{-1}$ (3). $-363 = 1920 + \Delta_{\text{LE}} \text{H} \Rightarrow \Delta_{\text{LE}} \text{H} = -363 - 1920 = -2283 \text{ kJ mol}^{-1}$ (2).                                                                                                                                                                                        | <b>[5]</b> |
| <b>Q18</b> | Q18: 18. Polarizing Power Comparison: $\text{Li}^+$ vs $\text{K}^+$<br>18. (a) $\text{Li}^+$ has a much smaller ionic radius than $\text{K}^+$ (1). $\text{Li}^+$ has higher charge density and higher polarizing power (1). $\text{Li}^+$ distorts the electron cloud of $\text{I}^-$ significantly more, introducing covalent character (1).                                                                                                                                                  | <b>[3]</b> |
| <b>Q19</b> | Q19: 19. Enthalpy of Hydration of Halide Ions<br>19. (a) Halide ionic radius increases down group ( $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$ ) (1). Charge density decreases down group (1). Electrostatic attraction between halide ion and $\delta^+$ hydrogen of water decreases (1). Hydration enthalpy becomes less exothermic (less negative) (1).                                                                                                                           | <b>[4]</b> |
| <b>Q20</b> | Q20: 20. Born-Haber Cycle for Aluminum Oxide $\text{Al}_2\text{O}_3$<br>20. (a) $\text{Al}^{3+}$ has high charge (+3) and small radius (1). $\text{O}^{2-}$ has high charge (-2) and small radius (1). Product of ionic charges is $3 \times 2 = 6$ for $\text{Al}_2\text{O}_3$ vs $1 \times 1 = 1$ for $\text{NaCl}$ (1). Electrostatic attraction between $\text{Al}^{3+}$ and $\text{O}^{2-}$ is immensely stronger (1). High charge density leads to extremely compact, stable lattice (1). | <b>[5]</b> |
| <b>Q21</b> | Q21: 21. Thermal Decomposition of Group 2 Nitrates<br>21. (a) $\text{Mg}^{2+}$ is smaller than $\text{Ba}^{2+}$ with higher charge density (1). $\text{Mg}^{2+}$ polarizes the nitrate $\text{N}-\text{O}$ bond more strongly (1). Weakens the $\text{N}-\text{O}$ bond in nitrate ion (1). Requires less thermal energy to break bond and decompose (1).                                                                                                                                       | <b>[4]</b> |
| <b>Q22</b> | Q22: 22. Solubility of Calcium Hydroxide vs Calcium Sulfate<br>22. (a) $\text{SO}_4^{2-}$ is much larger than $\text{OH}^-$ (1). $\text{CaSO}_4$ has larger lattice energy relative to hydration enthalpy (1). $\Delta_{\text{sol}} \text{H}$ for $\text{CaSO}_4$ is more endothermic (1). $\text{Ca}(\text{OH})_2$ has less endothermic $\Delta_{\text{sol}} \text{H}$ , allowing greater dissolution (1).                                                                                     | <b>[4]</b> |
| <b>Q23</b> | Q23: 23. Born-Haber Cycle for Lithium Iodide $\text{LiI}$<br>23. (a) $-270 = +159 + 520 + 107 - 295 + \Delta_{\text{LE}} \text{H} = +491 + \Delta_{\text{LE}} \text{H}$ (3). $\Delta_{\text{LE}} \text{H} = -270 - 491 = -761 \text{ kJ mol}^{-1}$ (1).<br>23. (b) Significant covalent character allows dissolution in non-polar/polar organic solvents (1).                                                                                                                                   | <b>[5]</b> |
| <b>Q24</b> | Q24: 24. Enthalpy of Hydration vs Lattice Energy for $\text{BaCl}_2$<br>24. (a) $\Delta_{\text{sol}} \text{H} = (-1305 + 2(-364)) - (-2056) = -2033 - (-2056) = -23 \text{ kJ mol}^{-1}$ (3).<br>24. (b) Exothermic ( $\Delta_{\text{sol}} \text{H} < 0$ ) (1).                                                                                                                                                                                                                                 | <b>[4]</b> |

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| <b>Q25</b> | Q25: 25. Born-Haber Cycle for Copper(II) Oxide CuO<br>25. (a) $\text{Sum} = +338 + 746 + 1958 + 249 + 657 = +3948 \text{ kJ mol}^{-1}$ (3). $-157 = 3948 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -157 - 3948 = -4105 \text{ kJ mol}^{-1}$ (2).                                                                                                                                                                                                                                                                                                                                                          | <b>[5]</b> |
| <b>Q26</b> | Q26: 26. A* Challenge: Born-Haber Cycle with Unknown Electron Affinity<br>26. (a) $-524 = 178 + 590 + 1145 + 112 + 2(\text{EA1}(\text{Br})) - 2150 = 2225 - 2150 + 2(\text{EA1}(\text{Br})) = +75 + 2(\text{EA1}(\text{Br}))$ (3). $2(\text{EA1}(\text{Br})) = -524 - 75 = -599 \Rightarrow \text{EA1}(\text{Br}) = -299.5 \text{ kJ mol}^{-1}$ (2).                                                                                                                                                                                                                                                                      | <b>[5]</b> |
| <b>Q27</b> | Q27: 27. A* Challenge: Percent Covalent Character from Lattice Energy Comparison<br>27. (a) $\% \text{ diff} = [(-905 - (-758)) / -758] \times 100\% = (-147 / -758) \times 100\% = 19.4\%$ (2).<br>27. (b) $\text{Ag}^+$ has 4d10 outer shell with incomplete shielding, high polarizing power (1). Large polarizable $\text{Br}^-$ electron cloud is distorted (1). Introduces substantial covalent orbital overlap, adding covalent bond energy to ionic attraction (1).                                                                                                                                               | <b>[5]</b> |
| <b>Q28</b> | Q28: 28. A* Challenge: Enthalpy of Solution Temperature Dependence & Entropy<br>28. (a) $\Delta_{\text{sol H}} = (-322 - 293) - (-649) = -615 - (-649) = +34 \text{ kJ mol}^{-1}$ (2).<br>28. (b) Dissolution is endothermic ( $\Delta_{\text{sol H}} > 0$ ). Increasing temperature shifts equilibrium $\text{KI(s)} \rightleftharpoons \text{K}^+(\text{aq}) + \text{I}^-(\text{aq})$ to the right (1). $\Delta S_{\text{surroundings}}$ becomes less negative, making $\Delta S_{\text{total}}$ more positive, increasing solubility (2).                                                                              | <b>[5]</b> |
| <b>Q29</b> | Q29: 29. A* Challenge: Lattice Energy of Hypothetical NaCl2<br>29. (a) $\sim -2500 \text{ kJ mol}^{-1}$ (1).<br>29. (b) $\text{IE2}(\text{Na})$ removes an electron from an inner 2p shell (noble gas core) requiring massive energy ( $+4562 \text{ kJ mol}^{-1}$ ) (2). This huge energy input outweighs the extra lattice energy gained by $\text{Na}^{2+}$ (1).<br>29. (c) $\Delta_{\text{f H}}(\text{NaCl2})$ is highly endothermic ( $+200$ to $+400 \text{ kJ mol}^{-1}$ ) whereas $\Delta_{\text{f H}}(\text{NaCl})$ is exothermic ( $-411 \text{ kJ mol}^{-1}$ ). $\text{NaCl}$ is thermodynamically stable (2). | <b>[6]</b> |
| <b>Q30</b> | Q30: 30. A* Challenge: Hydration Enthalpy vs Lattice Energy Competition in Halides<br>30. (a) As halide radius increases ( $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$ ), both lattice energy and hydration enthalpy become less exothermic (2). Lattice energy decreases much faster because $r_{\text{Li}^+} + r_{\text{halide}}$ decreases proportionally more for lattice packing (2). Decreasing lattice energy dominates over decreasing hydration enthalpy, making $\Delta_{\text{sol H}}$ increasingly exothermic (1).                                                                                  | <b>[5]</b> |
| <b>Q31</b> | Q31: 31. A* Challenge: Born-Haber Cycle for Calcium Oxide CaO<br>31. (a) $\text{Sum atom/IE/EA} = +178 + 590 + 1145 + 249 - 141 + 798 = +2819 \text{ kJ mol}^{-1}$ (2). $-635 = 2819 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -635 - 2819 = -3454 \text{ kJ mol}^{-1}$ (2).<br>31. (b) $2+$ and $2-$ charges in $\text{CaO}$ give 4x electrostatic attraction force compared to $1+$ and $1-$ in $\text{NaCl}$ (1).                                                                                                                                                                                      | <b>[5]</b> |
| <b>Q32</b> | Q32: 32. A* Challenge: Polarization Effect on Thermal Stability of Carbonates<br>32. (a) $\text{Na}^+$ has $+1$ charge (lower charge density) so it does not polarize the carbonate ion significantly (2). $\text{Ca}^{2+}$ and $\text{Mg}^{2+}$ have $+2$ charges and polarize carbonate C-O bond, weakening it (1).<br>32. (b) $\text{Mg}^{2+}$ is smaller than $\text{Ca}^{2+}$ so it has higher charge density and stronger polarizing power, weakening C-O bond more (2).                                                                                                                                            | <b>[5]</b> |
| <b>Q33</b> | Q33: 33. A* Challenge: Born-Haber Cycle for Barium Chloride BaCl2<br>33. (a) $\text{Sum} = 180 + 503 + 965 + 244 - 698 = +1194$ (2). $-858 = 1194 + \Delta_{\text{LE H}} \Rightarrow \Delta_{\text{LE H}} = -858 - 1194 = -2052 \text{ kJ mol}^{-1}$ (2).<br>33. (b) $\text{Ba}^{2+}$ has a larger ionic radius than $\text{Mg}^{2+}$ , leading to weaker electrostatic attraction and less exothermic lattice energy (1).                                                                                                                                                                                                | <b>[5]</b> |
| <b>Q34</b> | Q34: 34. A* Challenge: Hydration Enthalpy vs Ion Size in Transition Metal Ions<br>34. (a) $\text{Fe}^{3+}$ has higher charge ( $+3$ vs $+2$ ) and smaller ionic radius ( $0.64 \text{ \AA}$ vs $0.78 \text{ \AA}$ ) (1). $\text{Fe}^{3+}$ has much higher charge density, pulling water oxygen lone pairs much more strongly (2).<br>34. (b) $\text{Fe}^{3+}$ polarizes O-H bonds of attached water ligand molecules more strongly, weakening O-H bonds and releasing $\text{H}^+$ ions readily ( $\text{pH} \sim 2$ vs $\text{pH} \sim 5$ ) (2).                                                                         | <b>[5]</b> |
| <b>Q35</b> | Q35: 35. A* Challenge: Enthalpy Cycle for Anhydrous vs Hydrated Copper Sulfate<br>35. (a) $\Delta_{\text{H hyd}} = \Delta_{\text{sol H}}(\text{CuSO4(s)}) - \Delta_{\text{sol H}}(\text{CuSO4.5H2O(s)})$ (2) = $-66.5 - (+11.7) = -78.2 \text{ kJ mol}^{-1}$ (2).<br>35. (b) Water dissolves the solid as it hydrates, making direct measurement of solid hydration impossible (1).                                                                                                                                                                                                                                       | <b>[5]</b> |

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| <b>Q36</b> | Q36: 36. A* Challenge: Enthalpy of Solution of Potassium Nitrate KNO <sub>3</sub><br>36. (a) $\Delta_{\text{sol}} H = (-322 - 314) - (-687) = -636 - (-687) = +51.0 \text{ kJ mol}^{-1}$ (2).<br>36. (b) $\Delta S_{\text{surr}} = -51000 / 298 = -171.1 \text{ J K}^{-1} \text{ mol}^{-1}$ . $\Delta S_{\text{total}} = +115 - 171.1 = -56.1 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K. Dissolves at higher temperature as $\Delta S_{\text{surr}}$ becomes less negative (3).         | <b>[5]</b> |
| <b>Q37</b> | Q37: 37. A* Challenge: Born-Haber Cycle for Silver Chloride AgCl<br>37. (a) $-127 = +285 + 731 + 122 - 349 + \Delta_{\text{LE}} H = +789 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -127 - 789 = -916 \text{ kJ mol}^{-1}$ (3).<br>37. (b) Experimental value (-916) is 146 kJ mol <sup>-1</sup> more exothermic than theoretical (-770) due to significant covalent character / polarization of Cl <sup>-</sup> by Ag <sup>+</sup> (2).                                   | <b>[5]</b> |
| <b>Q38</b> | Q38: 38. A* Challenge: Enthalpy of Atomisation of Metals vs Lattice Energy<br>38. (a) Atomisation enthalpy increases from Na to Al (3).<br>38. (b) Cation charge increases (+1 to +3), ionic radius decreases, and number of delocalised electrons per atom increases (1 → 3), strengthening metallic bonding immensely (2).                                                                                                                                                              | <b>[5]</b> |
| <b>Q39</b> | Q39: 39. A* Challenge: Born-Haber Cycle for Zinc Sulfide ZnS<br>39. (a) $\text{Sum} = 131 + 2735 + 279 + 332 = +3477$ (2). $-206 = 3477 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -206 - 3477 = -3683 \text{ kJ mol}^{-1}$ (2).<br>39. (b) Zn <sup>2+</sup> strongly polarizes S <sup>2-</sup> (large polarizable anion), resulting in dominant covalent bonding character (1).                                                                                           | <b>[5]</b> |
| <b>Q40</b> | Q40: 40. A* Challenge: Dissolution Energetics of Calcium Chloride CaCl <sub>2</sub><br>40. (a) $\Delta_{\text{sol}} H = (-1577 + 2(-364)) - (-2258) = -2305 - (-2258) = -47 \text{ kJ mol}^{-1}$ (3).<br>40. (b) Dissolution is strongly exothermic ( $\Delta_{\text{sol}} H = -47 \text{ kJ mol}^{-1}$ ), generating heat (1). Dissolved ions depress freezing point of water effectively (1).                                                                                           | <b>[5]</b> |
| <b>Q41</b> | Q41: 41. A* Challenge: Polarization and Thermal Decomposition of Hydroxides<br>41. (a) Mg <sup>2+</sup> has a small ionic radius and high charge density / high polarizing power (1). Mg <sup>2+</sup> attracts oxygen electron density of the OH <sup>-</sup> ion (1). Weakens the O-H bond and facilitates splitting off H <sub>2</sub> O (1). Ba <sup>2+</sup> has a much larger radius and low charge density, so it cannot weaken O-H bond sufficiently to induce decomposition (2). | <b>[5]</b> |
| <b>Q42</b> | Q42: 42. A* Challenge: Enthalpy of Hydration of Gaseous Protons H <sup>+</sup> (g)<br>42. (a) H <sup>+</sup> is a bare proton with zero electron shell and extremely small radius (~10 <sup>-15</sup> m) (1). It has an infinitely high charge density (1), reacting covalently with water lone pair to form dative H <sub>3</sub> O <sup>+</sup> (aq) (1).<br>42. (b) $\text{H}^+(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq})$ (2).             | <b>[5]</b> |
| <b>Q43</b> | Q43: 43. A* Challenge: Born-Haber Cycle for Sodium Fluoride NaF<br>43. (a) $\text{Sum} = 107 + 496 + 79 - 328 = +354$ (2). $-574 = 354 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -574 - 354 = -928 \text{ kJ mol}^{-1}$ (2).<br>43. (b) F <sup>-</sup> is smaller than Cl <sup>-</sup> , allowing closer ionic approach and stronger electrostatic attraction (-928 vs -787) (1).                                                                                         | <b>[5]</b> |
| <b>Q44</b> | Q44: 44. A* Challenge: Solvation Enthalpies in Non-Aqueous Solvents<br>44. (a) NH <sub>3</sub> has a less electronegative nitrogen atom which donates its lone pair more readily to Li <sup>+</sup> (1). Forms stronger dative covalent coordination bonds [Li(NH <sub>3</sub> ) <sub>4</sub> ] <sup>+</sup> (2).<br>44. (b) Nitrogen is less electronegative than oxygen, holding its lone pair less tightly and forming stronger dative bonds to cations (2).                           | <b>[5]</b> |
| <b>Q45</b> | Q45: 45. A* Challenge: Theoretical vs Experimental Lattice Energies of Transition Metal Halides<br>45. (a) Cu <sup>2+</sup> (d <sup>9</sup> ) experiences strong Jahn-Teller distortion and high polarizing power, introducing significant covalent character (3).<br>45. (b) CFSE provides additional thermodynamic stabilization beyond simple electrostatic ionic model for d <sup>9</sup> Cu <sup>2+</sup> compared to high-spin d <sup>5</sup> Mn <sup>2+</sup> (CFSE = 0) (2).      | <b>[5]</b> |
| <b>Q46</b> | Q46: 46. A* Challenge: Born-Haber Cycle for Lead(II) Chloride PbCl <sub>2</sub><br>46. (a) $\text{Sum} = 195 + 2190 + 244 - 698 = +1931$ (2). $-359 = 1931 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -359 - 1931 = -2290 \text{ kJ mol}^{-1}$ (2).<br>46. (b) High covalent character and strong lattice energy make $\Delta_{\text{sol}} H$ endothermic (1).                                                                                                             | <b>[5]</b> |

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| <b>Q47</b> | Q47: 47. A* Challenge: Calculating Enthalpy of Solution from Calorimetry Data<br>47. (a) $q = 100 \times 4.18 \times 8.4 = 3511.2 \text{ J} = 3.511 \text{ kJ}$ (2).<br>47. (b) Moles = $5.00 / 111.0 = 0.04505 \text{ mol}$ (1). $\Delta_{\text{sol}} H = -3.511 / 0.04505 = -77.9 \text{ kJ mol}^{-1}$ (2).                                                                                                                                                                       | <b>[5]</b> |
| <b>Q48</b> | Q48: 48. A* Challenge: Born-Haber Cycle for Potassium Iodide KI<br>48. (a) Sum = $89 + 418 + 107 - 295 = +319$ (2). $-328 = 319 + \Delta_{\text{LE}} H \Rightarrow \Delta_{\text{LE}} H = -328 - 319 = -647 \text{ kJ mol}^{-1}$ (2).<br>48. (b) $\text{K}^+$ has low charge density (+1, large radius) so it does not polarize $\text{I}^-$ , resulting in almost 100% pure ionic bonding (1).                                                                                     | <b>[5]</b> |
| <b>Q49</b> | Q49: 49. A* Challenge: Hydration Enthalpy vs Ionic Size Grid Analysis<br>49. (a) $\text{K}^+ < \text{Na}^+ < \text{Ca}^{2+} < \text{Mg}^{2+}$ (1). $\text{K}^+$ is 1+ with largest radius (least exothermic) (1); $\text{Na}^+$ is 1+ with smaller radius (1); $\text{Ca}^{2+}$ is 2+ with larger radius than $\text{Mg}^{2+}$ (1); $\text{Mg}^{2+}$ is 2+ with smallest radius (highest charge density, most exothermic $\Delta_{\text{hyd}} H = -1920 \text{ kJ mol}^{-1}$ ) (1). | <b>[5]</b> |
| <b>Q50</b> | Q50: 50. A* Challenge: Complete Energetics Master Synthesis<br>50. (a) Sum = $148 + 2189 + 158 - 2(328) = 2495 - 656 = +1839$ (1.5). $\Delta_{\text{LE}} H = -1124 - 1839 = -2963 \text{ kJ mol}^{-1}$ (1.5).<br>50. (b) $\Delta_{\text{sol}} H = (-1920 + 2(-506)) - (-2963) = (-1920 - 1012) + 2963 = -2932 + 2963 = +31.0 \text{ kJ mol}^{-1}$ (3).                                                                                                                              | <b>[6]</b> |

**SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 12B****FAQ #1: Definition of Lattice Energy Direction****Category: Definition Trap****COMMON EXAMINER TRAP / MISTAKE:**

Defining lattice energy as dissociation (endothermic) instead of formation (exothermic).

**MODEL ANSWER / EXAMINER EXPECTATION:**

Edexcel IAL defines lattice energy as the enthalpy change when ONE mole of an ionic crystal is FORMED from its gaseous ions (always EXOTHERMIC, negative sign).

**FAQ #2: Multiplying Electron Affinity for Di-halides****Category: Stoichiometry Trap****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting to multiply EA1 by 2 for MX<sub>2</sub> salts like MgCl<sub>2</sub> or CaF<sub>2</sub>.

**MODEL ANSWER / EXAMINER EXPECTATION:**

For MX<sub>2</sub> compounds, 2 moles of halide ions are formed from 2 moles of halogen atoms. You MUST multiply EA1 by 2 (e.g.  $2 \times -349 = -698 \text{ kJ mol}^{-1}$ ).

**FAQ #3: Second Ionisation Energy for Group 2 Metals****Category: Ionisation Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including only IE1 for Group 2 metals in Born-Haber cycles.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Group 2 metals form M<sup>2+</sup> ions. The cycle MUST include both IE1 and IE2 (e.g.  $\text{Mg} \rightarrow \text{Mg}^+ \rightarrow \text{Mg}^{2+}$ ).

**FAQ #4: Sign of Second Electron Affinity****Category: EA2 Trap****COMMON EXAMINER TRAP / MISTAKE:**

Assuming EA2 is exothermic like EA1.

**MODEL ANSWER / EXAMINER EXPECTATION:**

EA1 is exothermic (attraction to neutral atom), but EA2 is ALWAYS ENDOTHERMIC (positive) due to electrostatic repulsion between incoming e<sup>-</sup> and 1<sup>-</sup> anion.

**FAQ #5: Theoretical vs Experimental Discrepancy****Category: Covalency Trap****COMMON EXAMINER TRAP / MISTAKE:**

Attributing lattice energy discrepancies to experimental measurement error.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Discrepancy occurs because theoretical model assumes 100% spherical ionic bonding. Experimental Born-Haber cycle includes COVALENT CHARACTER caused by cation polarization of anion.

**FAQ #6: Fajans' Polarization Criteria****Category: Fajans Rules****COMMON EXAMINER TRAP / MISTAKE:**

Confusing which ion polarizes and which is polarized.

**MODEL ANSWER / EXAMINER EXPECTATION:**

The CATION polarizes (requires high charge, small radius). The ANION is polarized (requires high charge, large polarizable electron cloud).

**FAQ #7: Hydration Enthalpy Sign in Solution Cycle****Category: Hess Cycle Trap****COMMON EXAMINER TRAP / MISTAKE:**

Subtracting hydration enthalpy instead of lattice energy in  $\Delta_{\text{sol}} H$ .

**MODEL ANSWER / EXAMINER EXPECTATION:**

$\Delta_{\text{sol}} H = \sum \Delta_{\text{hyd}} H(\text{ions}) - \Delta_{\text{LE}} H$ . Hydration is exothermic (-), lattice formation is exothermic (-), so subtracting lattice formation subtracts a negative number.

**FAQ #8: Group 2 Sulfate Solubility Trend****Category: Solubility Trend****COMMON EXAMINER TRAP / MISTAKE:**

Explaining sulfate solubility trend without emphasizing large sulfate ion size.

**MODEL ANSWER / EXAMINER EXPECTATION:**

$\text{SO}_4^{2-}$  is very large, so lattice energy changes little down group. Decreasing hydration enthalpy of cation dominates, making  $\Delta_{\text{sol}} H$  more endothermic down group (less soluble).

**FAQ #9: Group 2 Hydroxide Solubility Trend****Category: Solubility Trend****COMMON EXAMINER TRAP / MISTAKE:**

Confusing hydroxide solubility trend with sulfate trend.

**MODEL ANSWER / EXAMINER EXPECTATION:**

$\text{OH}^-$  is small, so lattice energy decreases sharply down group. Decreasing lattice energy dominates over decreasing hydration enthalpy, making  $\Delta_{\text{sol}} H$  more exothermic down group (more soluble).

**FAQ #10: Ionic Charge vs Radius Impact on Lattice Energy****Category: Lattice Comparison****COMMON EXAMINER TRAP / MISTAKE:**

Treating ionic radius as having equal weight to ionic charge.

**MODEL ANSWER / EXAMINER EXPECTATION:**

Ionic charge has a SQUARED effect on electrostatic attraction force ( $q_1 \times q_2$ ), so doubling charge quadruples lattice energy, whereas changing radius has a smaller percentage impact.